Conclusion

This final problem set wraps up some loose ends and points out a way forward.

1 Extra Problems

The following questions contain applied physics topics. They were not included in the main problem sets mostly because they used a variety of topics about equally, but they're great review practice.

[5] **Problem 1.** IPhO 2013, problem 2. A rather long but straightforward problem, with many numerical calculations, illustrating a real practical setup in detail. Use the data sheet of physical constants provided in the folder.

[5] **Problem 2.** IPhO 2013, problem 3. A solid question on an ice sheet, using thermodynamics and data analysis. Use the data sheet of physical constants provided in the folder.

[5] **Problem 3.** IZhO 2022, problem 2. A practical problem on estimating the greenhouse effect.

Solution. See the official solutions as usual. There's a minor typo in the marking scheme, as pointed out here.

[5] **Problem 4.** IPhO 2018, problem 3. Models in biophysics with fluids and thermodynamics.

[5] **Problem 5.** IPhO 2021, problem 1. The basics of geophysics, using fluids and waves.

[4] **Problem 6.** GPhO 2022, problem 2. Some simple estimates involving renewable energy.

Solution. See the official solutions as usual, and Stefan Ivanov's correction here.

[5] **Problem 7.** APhO 2014, problem 1. A challenging problem which builds intuition about atmospheric physics.

[5] **Problem 8.** APhO 2012, problem 3. A somewhat technical problem that combines polarization and interference, to derive a neat effect called a "geometric phase".

2 Extra Theoretical Exams

If you've made it this far, you're probably training for the IPhO. I've reserved questions from a few Olympiads you can take as full-length practice exams. The NBPhO is usually 2 days long, with 5 hours per day. However, it includes experimental questions that you can't do at home, and it's slightly easier than the IPhO, APhO, or EuPhO, so I've adjusted the time limits accordingly.

[8] **Problem 9.** USAPhO 2003.

[8] **Problem 10.** USAPhO 2005.

[10] **Problem 11.** USAPhO 2001.

[10] **Problem 12.** USAPhO 2000.

[10] **Problem 13.** NBPhO 2021.

[12] **Problem 14.** NBPhO 2019.

[12] **Problem 15.** GPhO 2019. (As noted here, there are several issues with the official solutions for problem 2.)

[15] **Problem 16.** IPhO 2015.

[15] **Problem 17.** NBPhO 2022. (For problems 7 and 8, see the errata here.)

[20] **Problem 18.** EuPhO 2020.

[20] **Problem 19.** EuPhO 2021.

[20] **Problem 20.** APhO 2017.

[20] **Problem 21.** IPhO 2019.

Remark

A few IPhOs have been omitted from the handouts entirely.

- IPhO 2017 has many errors. But it does illustrate some interesting ideas; if you want to try it, see the corrected version linked here.
- IPhO 2023 has many straightforward, standard subparts; its interesting new ideas have already been distributed throughout the handouts.
- IPhO 2024 is okay, though a bit straightforward and standard. Many of its core ideas overlap with other questions in the handouts. In particular, problem 1 is conceptually easy if you've done **T2**, but extremely messy and tedious.

Remark

I'm not a fan of the modern APhO. The problems tend to be split into a huge number of subparts. They are often unnecessarily tedious, with complicated algebraic expressions or lots of numbers to plug in. In addition, they often just ask the student to do standard manipulations covered in every 2nd or 3rd year college textbook, which means they don't test the more valuable skills of constructing a new argument from scratch, or deciding which kind of argument is right. Instead they just reward students who can be fast, reliable accounting machines. In particular, the following APhOs have been omitted from the handouts:

- APhO 2018 covers interesting applied physics, but it has 55 straightforward subparts.
- APhO 2019 is uneven in quality. Problem 1 is primarily hard because it uses a lot of jargon from more advanced physics. Problem 2 has major errors. Problem 3 is a good, but very tough 3D rotation problem.
- APhO 2020 was cancelled.

- APhO 2021 has the same issues as APhO 2018. Also, problem 1 is conceptually messy, and sensitive to modeling assumptions. Problem 2 is very tedious and well outside the Olympiad syllabus. Problem 3 is decent, though its solution is a bit confusing; it's included in **ERev**.
- APhO 2022 is okay, but easy; not much different from standard textbook exercises.
- APhO 2023 has some interesting ideas, but can be a bit tedious and ambiguous. If you want to try it, see Stefan Ivanov's errata first.
- APhO 2024 is extremely easy.

Once you finish this and get your IPhO medal, you've outgrown Olympiad physics. The stuff covered in Olympiads *is* real physics: the rough estimates and calculations a physicist would do when exploring a new idea actually resemble what you see in IPhO and APhO problems. But the IPhO and APhO are restricted by a syllabus that assumes very little mathematical background. There is a whole lot more beautiful, fascinating stuff you can do once you pick up vector calculus and linear algebra, the two most important pieces of the physicist's toolkit. For more advice on what you can do next, see my second advice file. You have a lot of options, and you've earned it.